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United States Patent	12384526
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Potier; Karl et al.

Spoiler actuator

Abstract

An electromechanical actuator comprises: a motor; a gearbox; an anti-extension device mounted between the motor and the gearbox. The motor and anti-extension device are mounted on a housing of the actuator. The actuator further comprises a screw shaft; a nut mounted on the screw shaft; an output rod connected to the nut, the output rod having a linear range of motion defined between full extension of the output rod at a first end point (A) and full retraction of the output rod at a second end point (C). The screw shaft, nut, and output rod are located in a cavity of the housing. The output rod and nut are held against rotation relative to the housing, such that rotation of the screw shaft drives the output rod to move linearly within the housing.

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Appl. No.:	18/348420
Filed:	July 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240034460 A1	Feb. 01, 2024

Foreign Application Priority Data

EP	22306118	Jul. 27, 2022
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Publication Classification

Int. Cl.: B64C13/50 (20060101); **B64C5/10** (20060101); **F16H25/20** (20060101); **F16H25/24** (20060101)

U.S. Cl.:

CPC B64C13/50 (20130101); **B64C5/10** (20130101); **F16H25/20** (20130101); **F16H25/2454** (20130101); **F16H2025/2081** (20130101)

Field of Classification Search

CPC: F16H (25/2021); F16H (25/2015); F16H (25/2454); F16H (2025/2081); B64C (13/50); B64C (13/506); B64C (13/343); B64C (13/38); B64C (13/26); B64C (13/30); B64C (13/32); B64C (13/34)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to European Patent Application No. 22306118.5 filed Jul. 27, 2022, the entire contents of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present document relates to an electromechanical actuator with an anti-extension device.

BACKGROUND

(3) Actuators are widely used in aircraft. In particular, electromechanical actuators (EMAs) are becoming more widely used for controlling the position of flaps, slats, a spoilers of aircraft. It is

known to locate anti-extension device between a gearbox and a motor of the actuator, wherein when the anti-extension device is activated, the motor is protected against torque that is feeding-back through the system, e.g. resulting from loads on the flap, slat, or spoiler. Weight considerations are often paramount in designing aircraft components and so designing lighter weight components is generally desired for aerospace applications.

(4) FIG. 1 shows a known design of electromechanical actuator (EMA) 10. The EMA 10 comprises a motor 12 that is connected, in sequence, to an anti-extension device 14, a torque-limiter 16, and to a gearbox 18. These components 12,14,16,18 are mounted to an external surface of a housing 20. The gearbox 18 extends through the housing and drives a screw shaft 22 in rotation. The screw shaft 22 extends within the housing 20 along an axis X. A nut 24 is mounted on the screw shaft and an output rod 26 is connected to the radially-outer side of the nut 24. The output rod 26 is held against rotation relative to the housing 20, e.g. via splines (not shown), and this holds the nut 24 against rotation relative to the housing 20. The output rod 26 has a connection end 28 for connection to a spoiler 34. The housing 20 has a housing connection end 30 whereby the housing 20 (and EMA 10 as a whole) may be connected to an airframe 32.

(5) In normal operation, the motor 12 rotates an output shaft (not shown) that turns the gears of the gearbox 18 which causes the screw shaft 22 to rotate within the housing 20. In some examples, the output shaft of the motor 20 may extend all the way through the anti-extension device 14 and into one side of the torque limiter 16. Another shaft (not shown) extends from the other side of the torque limiter 16 into the gearbox 18. In other examples, the anti-extension device 14 may contain its own shaft (e.g. shaft 56), and the motor output shaft connects to the anti-extension device's shaft 56 (e.g. via splines), and the anti-extension device's shaft 56 may connect to a first side of the torque limiter 16, so that torque from the motor 12 is transmitted to the gearbox.

(6) As the nut 24 is held against rotation relative to the housing 20, rotation of the screw shaft 22 within the housing 20 causes the nut 24 to translate linearly along the axis X, which thus causes the output rod 26 to translate linearly relative to the housing 20. In this manner, rotational output from the motor 22 is converted into a linear output of the EMA 10, by the output rod 26 moving in and out of the housing 20. When used on an aircraft (e.g. aircraft 33 shown in FIG. 9), this linear output of the EMA 10 controls the position of the spoiler 34 relative to the airframe 32.

(7) During flight, there may be significant aerodynamic forces on the spoiler 34. These forces may attempt to pull the output rod 26 further out from the housing 20, or push the output rod 26 into the housing 20. Due to the interaction of the nut 24 with the screw shaft 22, these linear forces on the output rod 26 are converted into torque in the screw shaft 22, which torque is transmitted, via the gearbox 18, into the torque limiter 16 and the anti-extension device 14. The anti-extension device 14 is designed to, when activated, react this torque and thereby prevent rotation of the screw shaft 22 that would allow the output rod 26 to extend further out of the housing 20. This operation is described in more detail below, in relation to FIG. 2.

(8) FIG. 2A shows a cross-section of the known anti-extension device 14. This cross-sectional drawing further shows the housing 20, upon which the anti-extension device 14 is mounted, and also shows the output rod 26 (inside the housing 20) in cross-section.

(9) The anti-extension device 14 comprises a solenoid 40 having a movable output 42 that bears against a first end of a pivotable rod 44. The pivotable rod 44 may be pivoted around a pivot point 46. A second end of the pivotable rod 44 abuts a pusher 48. The pusher 48 is biased by a spring 50. When the solenoid 40 is energized, the movable output 42 moves towards the pivotable rod 44 and pivots the rod 44 from a first position to a second position, said pivoting being about the pivot point 46 and against the bias from the spring 50.

(10) The pivotable rod 44 further comprises a pawl 52 that is arranged adjacent to a ratchet wheel 54. The ratchet wheel 54 is fixedly mounted on a shaft 56. When the anti-extension device 14 is connected to the motor 12, an output shaft of the motor 12 connects to the shaft 56 (or, as above, the shaft 56 may itself be the output shaft of the motor 12), such that torque from the motor 12

turns the shaft **56** and thus turns the ratchet wheel **54** if possible. On the other side of the anti-extension device, the shaft **56** also connects (or extends into) the torque limiter **16**.

(11) The pivotable rod **44** is arranged such that, in the first position (i.e. when the solenoid **40** is not energized, shown in FIG. 2A), the pawl **52** abuts the ratchet wheel **54** and, through engagement with teeth of the ratchet wheel **54**, prevents rotation of the ratchet wheel in one direction of rotation.

(12) FIG. 2B shows the second position (i.e. when the solenoid is energized), wherein the pivotable rod **44** has been pushed by the movable output **42** to a position where the pawl **52** does not abut the ratchet wheel **54**, and the ratchet wheel **54** and shaft **56** may rotate in either direction of rotation. The force from the movable output **42** to pivot the rod **44** also compresses the spring **50**. When the solenoid **40** is later de-energised, the force from the spring **50** pushes the pivotable rod **44** back to the first position, whereupon the pawl **52** reengages the ratchet wheel **54**.

(13) During operation of the EMA **10**, the motor **12** may be used to drive the output rod **26** in extension to a first desired position. The EMA **10** may then engage the anti-extension device by de-energising the solenoid **40**. This causes the pawl **52** to engage the ratchet wheel **54** and prevent rotation of the shaft **56** in a first direction of rotation. This first direction corresponds to extension of the output rod **26** out from the housing **20**. That is, when aerodynamic forces on the spoiler **34** seek to pull the output rod **26** further out from the housing **20**, this linear force is converted, by the interaction of the nut **24** with the screw shaft **22**, into a torque on the screw shaft **22**. This torque is then transmitted, via the gearbox **18** and torque limiter **16** into the anti-extension device **14**. There the torque is reacted by the engagement of the pawl **52** with the ratchet wheel **54**. This prevents the torque from being transmitted back into the motor **12** which could damage the motor **12**.

(14) If the torque on the shaft **56** in the anti-extension device **14** is too high when the solenoid **40** is energized, the frictional force between the ratchet wheel **54** and pawl **52** may be so great that the solenoid **40** cannot push the pivotable rod **44** so that the pawl **52** disengages from the ratchet wheel **54**. Typically during takeoff and landing, an aircraft spoiler may be extended to negative angular positions so that it follows an extended flap of the aircraft wing. This provides a largely continuous surface between a leading edge of the wing and the trailing edge of the flap. In the negative angular positions, the anti-extension mode may be active so that the spoiler cannot move from its negative angular position towards the neutral position/towards a positive position. However, if the pilot retracts the flap at this point, it is possible the flap will load against the extended spoiler, trying to push the extended spoiler back towards the spoiler's neutral/positive position. This force fight between the spoiler in the anti-extension mode and the flap could cause damage to either part. To avoid damage, the torque limiter **16** is provided. This device limits the maximum torque transmitted between the gearbox **18** and the anti-extension device **16**. The skilled reader will appreciate that, when the torque limit of the torque limiter **16** is exceeded, the torque limiter will allow rotation on the gearbox **18** side even while the shaft **56** on the side of the anti-extension device **14** is prevented from turning. This thus allows some rotation of the screw shaft **22** and, ultimately, extension of the spoiler **34** towards the neutral position.

SUMMARY

(15) According to a first aspect, there is provided an electromechanical actuator comprising: a motor; a gearbox; an anti-extension device mounted between the motor and the gearbox; wherein the motor and anti-extension device are mounted on a housing of the actuator; a screw shaft; a nut mounted on the screw shaft; an output rod connected to the nut, the output rod having a linear range of motion defined between full extension of the output rod at a first end point and full retraction of the output rod at a second end point; wherein the screw shaft, nut, and output rod are located in a cavity of the housing; wherein the output rod and nut are held against rotation relative to the housing, such that rotation of the screw shaft drives the output rod to move linearly within the housing; wherein the anti-extension device comprises: a ratchet wheel mounted on a shaft arranged to be driven by the motor; a pivotable rod comprising a pawl; the pivotable rod being

pivotal between a first position in which the pawl engages with the ratchet wheel to prevent rotation of the ratchet wheel in one direction of rotation, and a second position in which the pawl does not prevent rotation of the ratchet wheel; a cam connected to the pivotal rod, the cam movable between a first cam position and a second cam position, wherein the cam is biased by a spring towards the first cam position in which a portion of the cam extends through an aperture in the housing and into the cavity; wherein in the second cam position, the cam holds the pivotal rod at the second position in which the pawl does not prevent rotation of the ratchet wheel; wherein, when an axial end of the output rod is at a predetermined intermediate point or between the intermediate point and the second end point, the cam is moved to the second cam position by an outer surface of the output rod bearing against the cam; and wherein when the axial end of the output rod is between the intermediate point and the first end point, the output rod does not bear against the cam.

(16) In this manner, whenever the output rod is sufficiently retracted into the housing, the anti-extension device is automatically disabled such that the pawl does not engage with the ratchet wheel.

(17) The housing may comprise a housing connection end for connecting the actuator to an airframe; and the output rod may comprise an output rod connection end for connecting the actuator to a spoiler.

(18) The cam may have a curved or chamfered end; and/or the end of the output rod may be curved or chamfered.

(19) This may allow the cam to be moved smoothly into the anti-extension device when the output rod presses the cam thereinto.

(20) The anti-extension device may be directly connected to the gearbox. That is, the actuator may lack a torque limiter between the anti-extension device and the gearbox, for example.

(21) The anti-extension device may further comprise a solenoid having a movable output; wherein, when the axial end of the output rod is between the intermediate point and the first end point, movement of the movable output moves the pivotal rod to the second position.

(22) That is, when the output rod is not pressing the cam into the anti-extension device, the solenoid may be used to control the position of the pivotal rod and pawl relative to the ratchet wheel.

(23) In such examples, when an axial end of the output rod is at the predetermined intermediate point or between the intermediate point and the second end point, movement of the movable output does not affect the position of the pivotal rod.

(24) The electromechanical actuator may optionally not comprise a torque limiter at any location along a load path between the motor and the screw shaft. The existence of the cam and its interaction with the output rod and the anti-extension device may eliminate the need for any torque limiter, such as is often provided on prior art actuators.

(25) According to a second aspect, there is provided an aircraft comprising an airframe and a flight surface, and the electromechanical actuator as described hereinabove. In this aspect, the housing of the electromechanical actuator is connected to the airframe and the output rod of the electromechanical actuator is connected to the flight surface, such that the position of the output rod relative to the housing controls an angular position of the flight surface relative to the airframe.

(26) The actuator may be arranged such that when the end of the output rod is at a second intermediate axial point, the flight surface is at an angle of zero degrees relative to the airframe.

(27) The second intermediate axial point may be coincident with the predetermined intermediate point, such that the output rod bears against the cam only when the spoiler at an angle of greater than zero degrees relative to the airframe.

(28) The second intermediate point may be at a different axial location from the predetermined intermediate point. Optionally, the second intermediate point may be located axially between the

predetermined intermediate point and the second end point.

(29) The flight surface may be is a spoiler.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) Certain embodiments will now be described in greater detail by way of example only and with reference to the accompanying drawings in which:
- (2) FIG. 1 shows a known design of EMA for a spoiler, the EMA having an anti-extension device;
- (3) FIG. 2A shows a cross-section of the known EMA in the anti-extension mode;
- (4) FIG. 2B shows a cross-section of the known EMA in an active mode;
- (5) FIG. 3 shows a cross-section of a new design of EMA for a spoiler, the new EMA having a new design of anti-extension device;
- (6) FIG. 4 shows a cross-section of the new anti-extension device;
- (7) FIG. 5 shows a second cross-section of the new EMA;
- (8) FIG. 6 shows a second cross-section of the new anti-extension device;
- (9) FIG. 7 shows positions of the spoiler controlled by the EMA;
- (10) FIG. 8 shows various end profiles of a cam and of an output rod of the new EMA; and
- (11) FIG. 9 shows an aircraft.

DETAILED DESCRIPTION

- (12) A new design of EMA **100** will now be described in relation to FIGS. 3 to 8. The new design of EMA **100** shares several similarities with the known EMA **10** design described hereinabove, and where like components are used, the same reference numeral will be used.
- (13) The EMA **100** shown in FIG. 3 comprises a motor **12** and a new design of anti-extension device **140** having a cam **142**. The anti-extension device **140** connects directly between the motor **12** and a gearbox **18**. That is, compares to the known EMA **10** described above, there is no torque-limiter **16** present in the new design of EMA **100**. Removal of the torque limiter may reduce the overall weight of the new design of EMA **100** compared to the known EMA **10** described above.
- (14) The motor **12**, anti-extension device **140**, and gearbox **18** are mounted to a housing **200**. In one example, the housing **200** may be identical to the housing **20** of FIG. 1 except that the housing **200** comprises an aperture **202** through which the cam **142** extends. Thus, a housing **20** from the known design of EMA **10** may be simply converted into the new design of housing **200** simply by drilling a hole through the housing **20**, to form the aperture **202** that receives the cam **142**. The aperture **202** allows the cam **142** to extend from the anti-extension device **140**, through the housing **200**, and into a cavity **204** inside the housing, in which the output rod **26** moves.
- (15) As described in greater detail below, the cam **142** is movable along a second axis Y between a first cam position (shown in FIG. 4) and a second cam position (shown in FIG. 6). The second axis Y is non-parallel with the first axis X and may be generally perpendicular to the first axis X.
- (16) The output rod **26** has a connection end **28** for connection to a spoiler **34** and the housing **200** has a housing connection end **30** where the housing (and EMA **100** as a whole) may be connected to the airframe **32**. When the cam **142** is pressed in to the anti-extension device **140**, the anti-extension functionality of the anti-extension device **140** is automatically disabled, i.e. the anti-extension device **140** does not prevent extension of the output rod **26** while the anti-extension device is disabled, regardless of whether the solenoid **40** is energized or not, in the manner discussed in greater detail below.
- (17) As before, the gearbox **18** transmits torque from the motor **12** to turn the screw shaft **22**. The nut **24** is mounted on the screw shaft **22** and the nut connects to the output rod **26** which, like before, is held against rotation relative to the housing **200**.
- (18) During normal operation, when the motor **12** of FIG. 3 is activated, torque is transmitted

through the anti-extension device **140** and gearbox **18**, to turn the screw shaft **22**. Rotation of the screw shaft causes the nut **24**, and therefore the output rod **26**, to move linearly relative to the housing **200**. As such, rotational output from the motor **12** is, as before, converted into linear movement of the output rod **26** and this linear movement of the output rod **26** sets the position of the spoiler **34** relative to the airframe **32**.

(19) FIG. **4** shows a cross-sectional view of the new design of anti-extension device **140**. The new anti-extension device **14** shares a number of similar components to the anti-extension device **14** described above in relation to FIG. **2**, and where like components are used, the same reference numeral will be used.

(20) The anti-extension device **140** comprises the solenoid **40** having the movable output **42** that bears against the first end of a pivotable rod **44**. The pivotable rod **44** may be pivoted around the pivot point **46**.

(21) While the cam **142** is not pushed into the anti-extension device **140** by the output rod **26**, the anti-extension device is not disabled. When the anti-extension device is not disabled, the position of the pawl **52** with respect to the ratchet wheel **54** is under the control of the solenoid **40**. That is, when the solenoid **40** is energized, the movable output moves towards the pivotable rod **44** and pivots the rod, from a first rod position to a second rod position, said pivoting being about the pivot point **46** and against the bias from the spring **50**. When the solenoid is de-energized, the bias from the spring **50** pivots the pivotable rod back to its initial first rod position. As such, while the cam **142** is not being pushed into the anti-extension device **140** by the output rod **26**, the solenoid **40** controls whether the anti-extension device acts to prevent extension of the output rod **26** or not.

(22) A second end of the pivotable rod **44** abuts the cam **142**. The cam **142** comprises a first flange **144** and a second flange **146**. The spring **50** biases the cam **142** outward from the anti-extension device **140** (and therefore, biases the cam **142** generally in towards the axis X, into the cavity **204** of the housing **200**)—this position is the first cam position of the cam **142**. The second end of the pivotable rod **44** is located between the first and second flanges **144,146** of the cam **142**, such that movement of the cam **142** in either direction along the second axis Y controls the angular position of the pivotable rod **44** about the pivot point **46**. That is, when the cam **142** moves in to the anti-extension device **140**, to its second cam position, this movement causes the second flange **146** to bear against the second end of the pivotable rod **44** and pivot the rod **44** such that the pawl **52** disengages the ratchet wheel **54**. Movement of the cam **142** further out from the anti-extension device **140**, under bias from the spring **50**, causes the first flange **144** to bear against the pivotable rod **44** and pivot the rod **44** so as to bring the pawl **52** back into engagement with the ratchet wheel **54**. The cam **142** may only move out from the anti-extension device **140**, under bias from the spring **50**, at times when the output rod **26** is not pressing the cam **142** into the anti-extension device **140**.

(23) At times when the output rod **26** is not pressing the cam **142** into the anti-extension device, the pivotable rod **44** may be moved by the solenoid **40** such that the pawl **52** is disengaged from the ratchet wheel **54**. When the solenoid **40** moves the pivotable rod **44** to such a disengaged rod position, the second end of the pivotable rod **44** presses against the first flange **144** and, against the bias of the spring **50**, pulls the cam **142** into the anti-extension device **140**.

(24) The output rod **26** is linearly movable relative to the housing **200** along a range of motion depicted by the line A-C in FIG. **3**. Point A corresponds to full extension of the output rod **26**, i.e. out from the housing **200**, and point C corresponds to full retraction of the output rod **26**. More precisely, point A may correspond to a first axial position of an end **26b** of the output rod **26** within the housing **200**, and point C may correspond to a second axial position of the end **26b** of the output rod **26** within the housing **200**.

(25) When connected to an airframe **32** and spoiler **34**, the linear position of the output rod **26** relative to the housing **200** sets the angle of the spoiler **34** relative to the airframe **32**. Various angles of the spoiler **34** are shown in FIG. **7**. The EMA **100** may be arranged such that a given axial

position B of the output rod **26** within the housing **200** corresponds to a 0° position of the spoiler **34**. The position B may be located between the ends A and C of the range of motion available to the output rod **26**. That is, the EMA **100** may be arranged to allow the spoiler **34** to be moved within a range of positive angles, a range of negative angles, and to be held at a 0° position. For example, the spoiler **34** may be movable between $+15^\circ$ and -50° , where $+15^\circ$ of the spoiler **34** corresponds to the output rod **26** being at full retraction at position C, and where -50° of the spoiler **34** corresponds to the output rod **26** being at full extension at position A.

(26) When the spoiler **34** is at positive angular positions, i.e. its angular position is $>0^\circ$, the spoiler **34** is said to be operating in the “droop area”. The droop area of the spoiler **34** thus corresponds to the output rod **26** being in the linear range of $<B$ and $\geq C$ relative to the housing **200**.

(27) In the position depicted in FIG. 3, the output rod **26** is at a linear position within the cavity **204** where the output rod **26** does not abut against the cam **142**. When the output rod **26** retracts within the cavity **204**, it will eventually come into contact with an angled end **148** of the cam **142**. The angled end **148** may be a curved end, e.g. hemispherical, or may be e.g. chamfered (as shown in FIG. 8).

(28) The angled end **148** allows the cam **142** to ride up onto a radially outer surface **26a** of the output rod **26** when, during retraction of the output rod **26**, an end **26b** of the output rod **26** first comes into alignment with the cam **142**. The end **26b** of the output rod **26** may also be chamfered or curved (as shown in FIG. 8) to allow a smooth motion of pushing the cam **142** out from the cavity **204**. With brief reference to FIG. 5, this Figure depicts a cross-section of the EMA **100** where the output rod **26** is retracted sufficiently far into the housing **200** that the cam **142** is pressed away from the first axis X and into the anti-extension device **140**.

(29) An intermediate axial position D of the output rod **26**, between the end points A and C, is defined as the point where the end **26b** of the output rod **26** first bears against the angled end **148** of the cam **142** and has pushed the cam **142** into the anti-extension device **140** sufficiently far so as to cause the pawl **52** to disengage from ratchet wheel **54**. When the output rod **26** is pressing the cam **142** into the anti-extension device **140**, the anti-extension device **140** is disabled. That is, in this cam position, regardless of whether the solenoid **40** is energized or not, the pawl **52** does not contact the ratchet wheel **54** and thus the shaft **56** is free to rotate in either direction.

(30) Intermediate axial position D (where the end **26b** of the output rod **26** first meets the cam **142**) may be identical to position B (where the spoiler is at) 0° , but this is not essential. When axial positions D and B are identical, i.e. at the same axial location along axis X, this means that the anti-extension device **140** is disabled whenever the spoiler **34** is at 0° or higher. In this example, when the spoiler **34** is at angles $<0^\circ$, the anti-extension device **140** is no longer disabled and the engagement/disengagement of the pawl **52** from the ratchet wheel **54** is again under the control of the solenoid **40**.

(31) In other embodiments, intermediate axial position D is offset from position B. For example, the intermediate axial position D may be set at a spoiler angle of $+10^\circ$. In this example, the anti-extension device **140** will be disabled whenever the spoiler **34** is at a 10° angle or higher.

(32) FIG. 6 shows the cross-section of the anti-extension device **140** when the output rod **26** is at the rod position shown in FIG. 5, i.e. where the output rod **26** has pushed the cam **142** in to the anti-extension device **140**. Here, the first flange **144** has compressed the spring and the second flange **146** has pushed the second end of the pivotable rod **44** to the second rod position wherein the pawl **52** does not engage the ratchet wheel **54**. As may be seen, the movable output **42** of the solenoid **40** is still in its original position (i.e. the solenoid **40** is not energized) and the movable output **42** is not presently in abutment with the pivotable rod **44**. Thus, when the output rod **26** has pushed the cam **142** into anti-extension device **140**, the pawl **52** is disengaged from the ratchet wheel **52** regardless of whether or not the solenoid **40** is energized. That is, the anti-extension device **140** is automatically disabled whenever the output rod **26** is pressing the cam **142** into the anti-extension device **140**.

(33) When the output rod **26** extends beyond position D, the cam **142** is biased back into the cavity **204** by the force of the spring **50**. At this point, the anti-extension device **140** is again under control of the solenoid **40**. That is, if the solenoid **40** is now energized, the movable output **42** will push the pivotable rod **44** to the second position, where the pawl **52** is disengaged from the ratchet wheel **54**.

(34) The skilled reader will appreciate that, when the (end **26b** of the) output rod **26** is anywhere within the linear range between points A and D, the solenoid **40** determines whether the anti-extension device **140** is engaged or not. When the (end **26b**) of the output rod **26** is anywhere within the linear range between positions D and C, the anti-extension device **140** is automatically disabled, by virtue of the cam **142** being pushed into the anti-extension device **140** and thereby disengaging the pawl **52** from the ratchet wheel **54**.

(35) FIG. **8** depicts various alternative shapes for the angled end **148** of the cam **142** and for the end **26b** of the output rod **26**. FIG. **8** shows: a chamfered end of the output rod **26**; a rounded end of the output rod **26**; and a chamfered end of the cam **142**. The EMA **100** may use any combination of the various end shapes depicted in any of the Figures for the cam **142** and for the output rod **26**.

(36) FIG. **9** shows an aircraft **33** having the spoiler **34**. The aircraft **33** comprises the airframe **32** and the EMA may be connected to any suitable part of the airframe, such as inside of the wing or on the underside of the wing, near the spoiler **34**.

(37) While the foregoing discussion has discussed control of a spoiler **34**, the electromechanical actuator **100** may be used to control different control surfaces of an aircraft, including flaps, slats, ailerons etc.

(38) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(39) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. An electromechanical actuator comprising: a housing of the actuator; a motor; a gearbox; an anti-extension device mounted between the motor and the gearbox, wherein the motor and anti-extension device are mounted on the housing of the actuator; a screw shaft; a nut mounted on the screw shaft; and an output rod connected to the nut, the output rod having a linear range of motion defined between full extension of the output rod at a first end point (A) and full retraction of the output rod at a second end point (C); wherein the screw shaft, nut, and output rod are located in a cavity of the housing; wherein the output rod and nut are held against rotation relative to the housing, such that rotation of the screw shaft drives the output rod to move linearly within the housing; wherein the anti-extension device comprises: a ratchet wheel mounted on a shaft arranged to be driven by the motor; a pivotable rod comprising a pawl; the pivotable rod being pivotable between a first rod position in which the pawl engages with the ratchet wheel to prevent rotation of the ratchet wheel in one direction of rotation, and a second rod position in which the pawl does not

prevent rotation of the ratchet wheel; and a cam connected to the pivotable rod, the cam movable between a first cam position and a second cam position, wherein the cam is biased by a spring towards the first cam position in which a portion of the cam extends through an aperture in the housing and into the cavity; wherein, in the second cam position, the cam holds the pivotable rod at the second rod position in which the pawl does not prevent rotation of the ratchet wheel; wherein, when an axial end of the output rod is at a predetermined intermediate point (D) or between the intermediate point (D) and the second end point (C), the cam is moved to the second cam position by an outer surface of the output rod bearing against the cam; and wherein when the axial end of the output rod is between the intermediate point (D) and the first end point (A), the output rod does not bear against the cam.

2. The electromechanical actuator of claim 1, wherein the housing comprises a housing connection end for connecting the actuator to an airframe; and wherein the output rod comprises an output rod connection end for connecting the actuator to a spoiler.
 3. The electromechanical actuator of claim 1, wherein the cam has a curved or chamfered end; or wherein the end of the output rod is curved or chamfered.
 4. The electromechanical actuator of claim 1, wherein the anti-extension device is directly connected to the gearbox.
 5. The electromechanical actuator of claim 1, wherein the anti-extension device further comprises a solenoid having a movable output; wherein, when the axial end of the output rod is between the intermediate point (D) and the first end point (A), movement of the movable output moves the pivotable rod to the second rod position.
 6. The electromechanical actuator of claim 5, wherein when an axial end of the output rod is at the predetermined intermediate point (D) or between the intermediate point (D) and the second end point (C), movement of the movable output does not affect the position of the pivotable rod.
 7. The electromechanical actuator of claim 1, wherein the electromechanical actuator does not comprise a torque limiter at any location along a load path between the motor and the screw shaft.
 8. An aircraft comprising: an airframe; a flight surface; and an electromechanical actuator as recited in claim 1; wherein the housing of the electromechanical actuator is connected to the airframe and the output rod of the electromechanical actuator is connected to the flight surface, such that the position of the output rod relative to the housing controls an angular position of the flight surface relative to the airframe.
 9. The aircraft of claim 8, wherein the actuator is arranged such that when the end of the output rod is at a second intermediate axial point (B), the flight surface is at an angle of zero degrees relative to the airframe.
 10. The aircraft of claim 9, wherein the second intermediate axial point (B) is coincident with the predetermined intermediate point (D), such that the output rod bears against the cam only when the spoiler at an angle of greater than zero degrees relative to the airframe.
 11. The aircraft of claim 9, wherein the second intermediate point (B) is at a different axial location from the predetermined intermediate point (D).
 12. The aircraft of claim 11, wherein the flight surface is a spoiler.
 13. The aircraft of claim 11, wherein the second intermediate point is located axially between the predetermined intermediate point (D) and the second end point (C).
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